

Technology Stack

Date	25 September 2026
Team ID	6ab22227a614981d6d2c6451
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

Table-1: Components & Technologies:

S.N	Component	Description	Technology
1.	User Interface	How the user interacts with the application to upload files and view dashboards.	HTML, CSS, JavaScript (ServiceNow Standard UI / Next Experience).
2.	Application Logic-1	Logic for loading and staging the data from Excel.	ServiceNow Import Set API.
3.	Application Logic-2	Logic for transforming and mapping the data.	ServiceNow Transform Engine (Rhino JavaScript / GlideRecord).
4.	Database	Relational data tables holding Employee records.	ServiceNow Relational DB (MariaDB/MySQL backend).
5.	Cloud Infrastructure	Deployment environment and hosting.	ServiceNow Platform-as-a-Service (PaaS).
6.	File Storage	Processing the uploaded .xlsx files.	ServiceNow Attachment Storage.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks utilized within the platform environment.	Glide API, Rhino JavaScript.
2.	Security Implementations	Access controls and data security.	ServiceNow ACLs (Access Control Lists), Role-based access, HTTPS.
3.	Scalable Architecture	Justify scalability of the application.	ServiceNow Multi-Instance Architecture (isolates data for high scalability).
4.	Availability	Justify the availability of the application.	ServiceNow Advanced High Availability (AHA) architecture with automated backups.
5.	Performance	Design considerations for fast data imports.	Database indexing on the "Coalesce" field (Employee ID) to optimize queries.